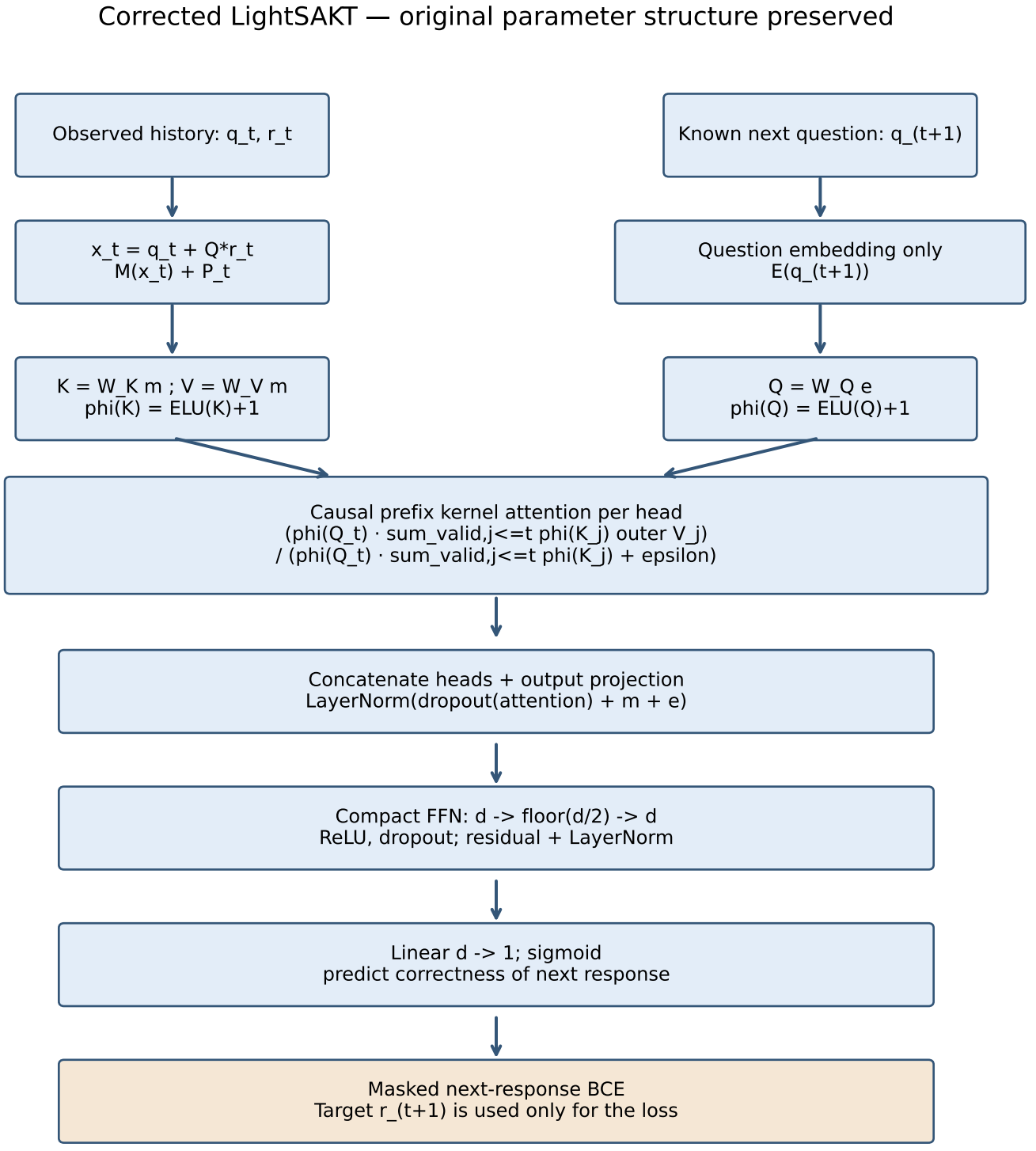
Corrected LightSAKT — original parameter structure preserved
Observed history: q_t, r_t
Known next question: q_(t+1)
x_t = q_t + Q*r_t
M(x_t) + P_t
Question embedding only
E(q_(t+1))
K = W_K m ; V = W_V m
phi(K) = ELU(K)+1
Q = W_Q e
phi(Q) = ELU(Q)+1
Causal prefix kernel attention per head
(phi(Q_t) · sum_valid,j<=t phi(K_j) outer V_j)
/ (phi(Q_t) · sum_valid,j<=t phi(K_j) + epsilon)
Concatenate heads + output projection
LayerNorm(dropout(attention) + m + e)
Compact FFN: d -> floor(d/2) -> d
ReLU, dropout; residual + LayerNorm
Linear d -> 1; sigmoid
predict correctness of next response
Masked next-response BCE
Target r_(t+1) is used only for the loss